

JavaScript Revision Notes (Professional Class Handout) Roman Urdu Version

1. Array Basic & Higher Order Methods

Array JavaScript ka built-in data structure hota hai jisme multiple values store hoti hain. Basic methods jaise push, pop, shift, unshift data ko add/remove karte hain. Higher order methods jaise map, filter, reduce, forEach functions ko argument ke tor par accept karte hain aur real-world projects mai data transformation, filtering aur calculations ke liye use hotay hain.

2. Prototype Based Inheritance

JavaScript prototype based language hai. Har object ka ek prototype hota hai jahan se wo properties aur methods inherit karta hai. Is system se code reuse hota hai aur duplicate logic likhne ki zarurat nahi hoti. Ye inheritance ka JS style hai jo classical inheritance se different hota hai.

3. Closures & Currying

Closure ka matlab hota hai ke inner function apne outer function ke variables ko access kar sakta hai chahe outer function execute ho chuka ho. Currying ka matlab hota hai function ko multiple small functions mai divide karna jahan har function ek argument receive karta hai. Ye functional programming concept hai.

4. Local Storage

Local Storage browser ki permanent storage hoti hai jahan data key-value pair mai save hota hai. Browser close hone ke baad bhi data safe rehta hai jab tak manually clear na kiya jaye. Mostly authentication, cart system aur user preferences ke liye use hoti hai.

5. API (Application Programming Interface)

API systems ke darmiyan communication ka zariya hoti hai. JavaScript mai API ka use backend se data fetch karne ke liye hota hai jaise Fetch API, Axios waghera. API real-time applications ki backbone hoti hai.

6. Destructuring, Rest & Spread Operators

Destructuring ka matlab object ya array se direct values extract karna. Rest operator (...) multiple values ko ek array mai collect karta hai. Spread operator (...) arrays/objects ko copy, merge aur clone karne ke kaam aata hai.

7. Call, Bind, Apply

Call, bind aur apply functions ka use this context change karne ke liye hota hai. Call aur apply function ko immediately execute kar dete hain jabke bind naya function return karta hai jo baad mai execute hota hai. Ye advanced function handling concepts hain.

8. OOP with JavaScript

OOP ka matlab Object Oriented Programming hota hai. JavaScript mai objects, classes, inheritance, encapsulation, abstraction aur polymorphism use hota hai. OOP ka purpose scalable, reusable aur maintainable code likhna hota hai.